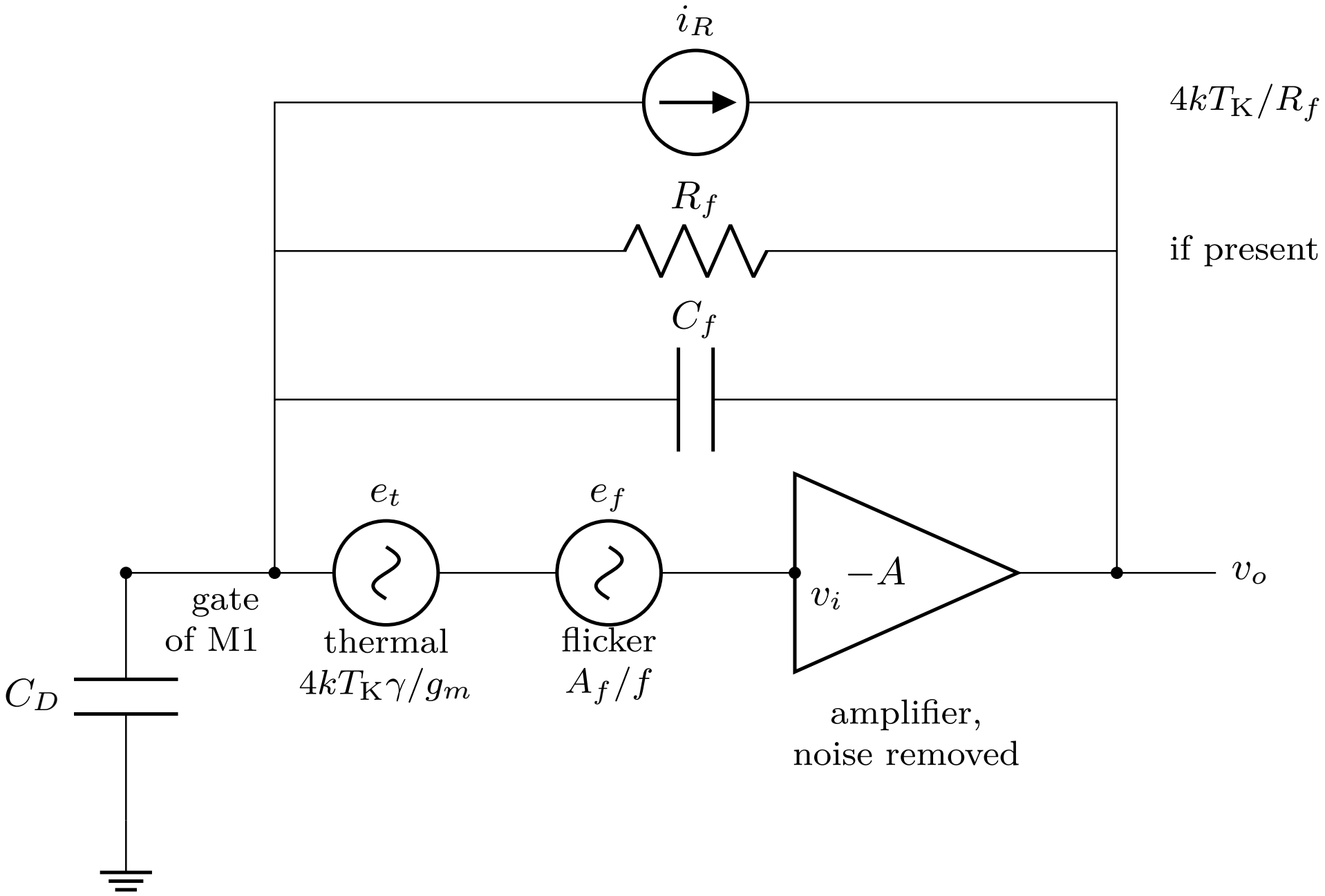

$i_R$
$4kT_K/R_f$
$R_f$
if present
$C_f$
$e_t$
$e_f$
$-A$
$v_i$
$v_o$
gate
of M1
thermal
$4kT_K\gamma/g_m$
flicker
$A_f/f$
$C_D$
amplifier,
noise removed